Bitcoin



Networks

- **Mainnet.** The Bitcoin mainnet refers to the original and official network of Bitcoin. It is the production version of the Bitcoin blockchain, where real transactions take place and Bitcoin is mined.
- **Testnet.** The Bitcoin testnet is a separate network that allows developers, researchers, and users to experiment and test new features, applications, or upgrades without risking real bitcoins or affecting the mainnet.
- **Regtest.** The Bitcoin regtest, short for "Regression Test Mode," is a mode available in Bitcoin software that allows developers and users to create a local private blockchain for testing and development purposes.

Bitcoin Core

```
docker run --name=bitcoind-node -d \
  --mount type=bind,source=/d/bitcoin,target=/bitcoin/.bitcoin \
  -p 18332:18332 \
  kylemanna/bitcoind
```

Approximate mainnet size = 600 Gb

Aproximate testnet size = 35 Gb

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Bitcoin Core Configuration

Configuration tool

https://jlopp.github.io/bitcoin-core-config-generator/

```
# Generated by https://jlopp.github.io/bitc

# [rpc]
# Accept command line and JSON-RPC commands
server=1

# [core]
# Maintain a full transaction index, used b
txindex=1

# Run this node on the Bitcoin Test Network
testnet=1

rpcuser=user
rpcpassword=password

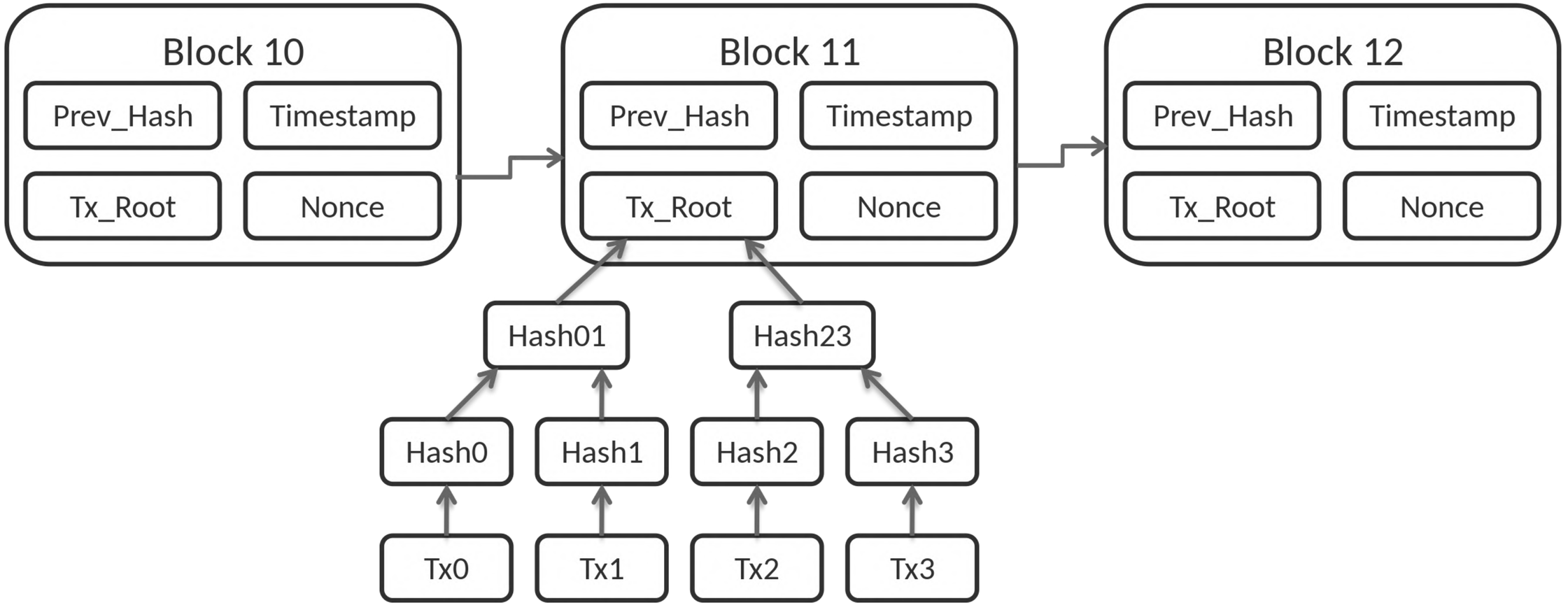
# [Sections]
# Most options automatically apply to mainn
# If you want to confine an option to just
# EXCEPTIONS: The options addnode, connect,
# only apply to mainnet unless they appear

[test]

# Bind to given address to listen for JSON-
rpcbind=0.0.0.0
# Allow JSON-RPC connections from specifiea
rpcallowip=0.0.0.0/0
```

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Blocks


Block 10
Prev_Hash
Timestamp
Tx_Root
Nonce
Block 11
Prev_Hash
Timestamp
Tx_Root
Nonce
Block 12
Prev_Hash
Timestamp
Tx_Root
Nonce
Hash01
Hash23
Hash0
Hash1
Hash2
Hash3
Tx0
Tx1
Tx2
Tx3

Asymmetric Encryption

Asymmetric encryption, also known as public-key cryptography, is a cryptographic system that uses two different but mathematically related keys for encryption and decryption. This approach provides a higher level of security compared to symmetric encryption, where the same key is used for both encryption and decryption.

In asymmetric encryption, there are two keys: a public key and a private key. The public key is widely distributed and can be freely shared with anyone, while the private key is kept secret and known only to the owner.

When someone wants to send an encrypted message to a recipient, they use the recipient's public key to encrypt the message. Once encrypted, the message can only be decrypted using the corresponding private key, which is held by the recipient.

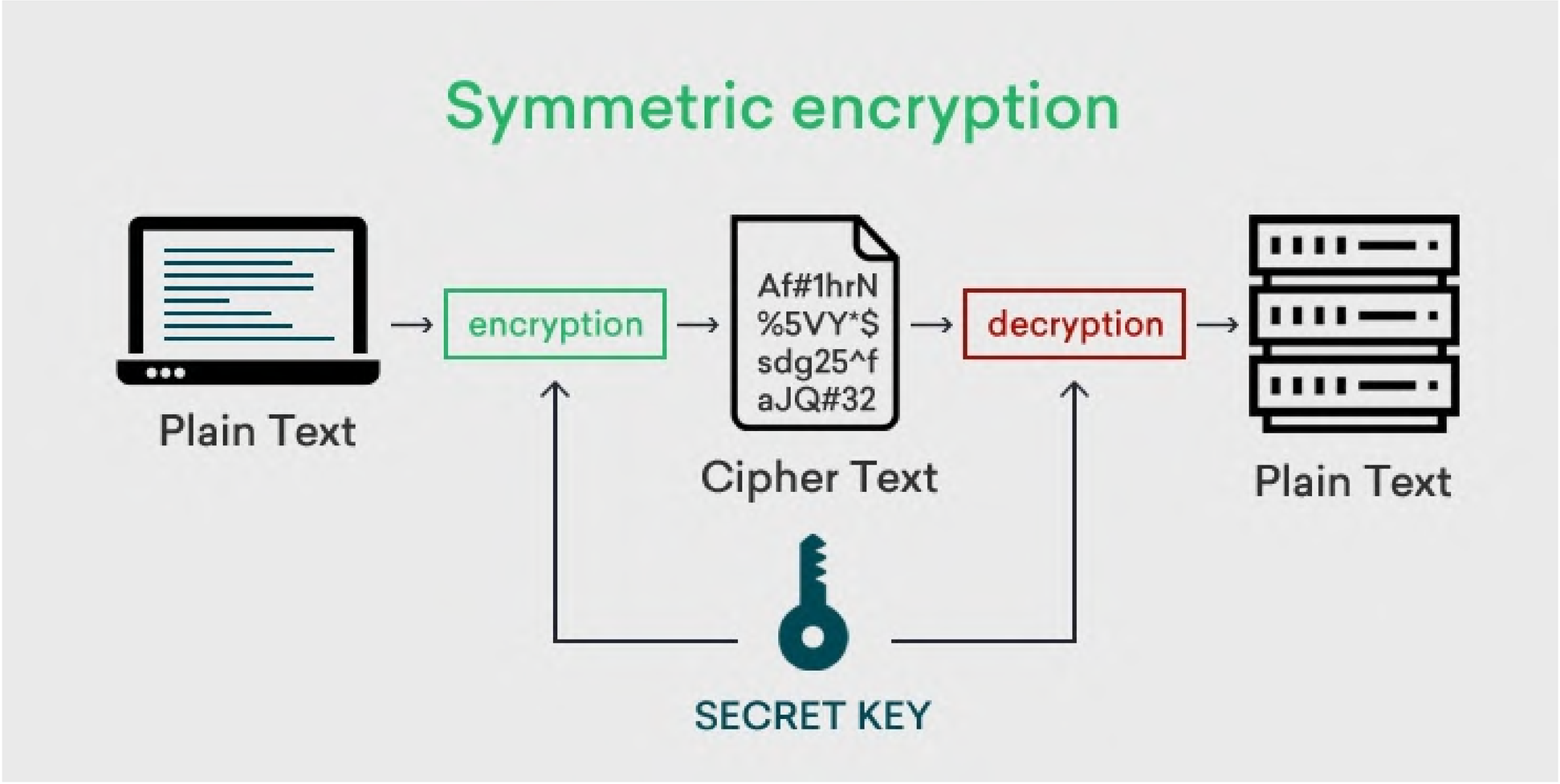
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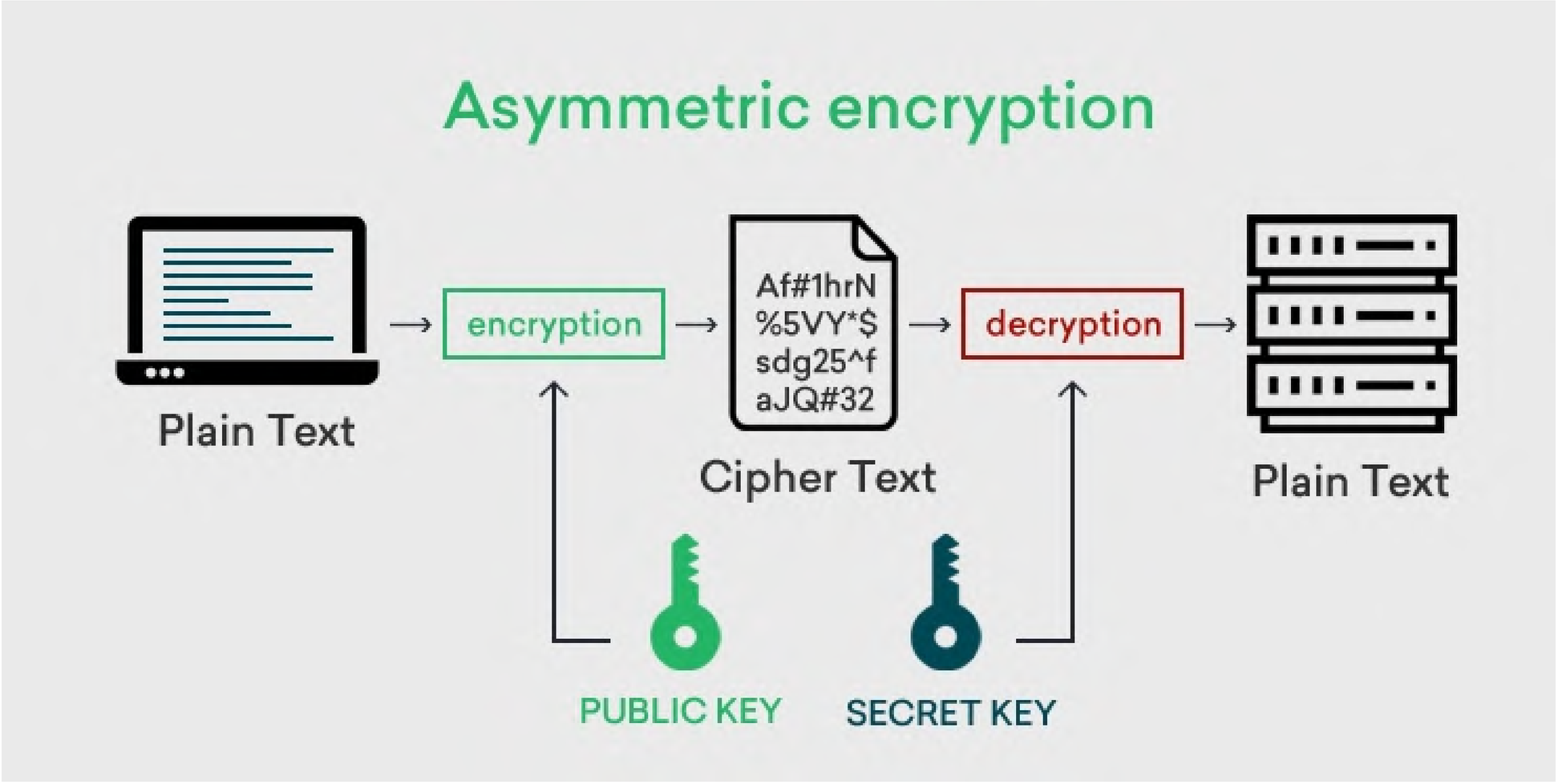
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Symmetric Encryption



Asymmetric Encryption



Symmetric Encryption Algorithms

- AES (Advanced Encryption Standard)
- DES (Data Encryption Standard)
- Blowfish
- Twofish

Asymmetric Encryption Algorithms

- RSA (Rivest–Shamir–Adleman)
- DH (Diffie–Hellman)
- ECC (Elliptic Curve Cryptography)
- ElGamal

Bitcoin uses ECC.

Addresses

ECC Public Key

`025c0de3b9c8ab18dd04e3511243ec2952002dbfadc864b9628910169d9b9b00ec`

ECC Private Key

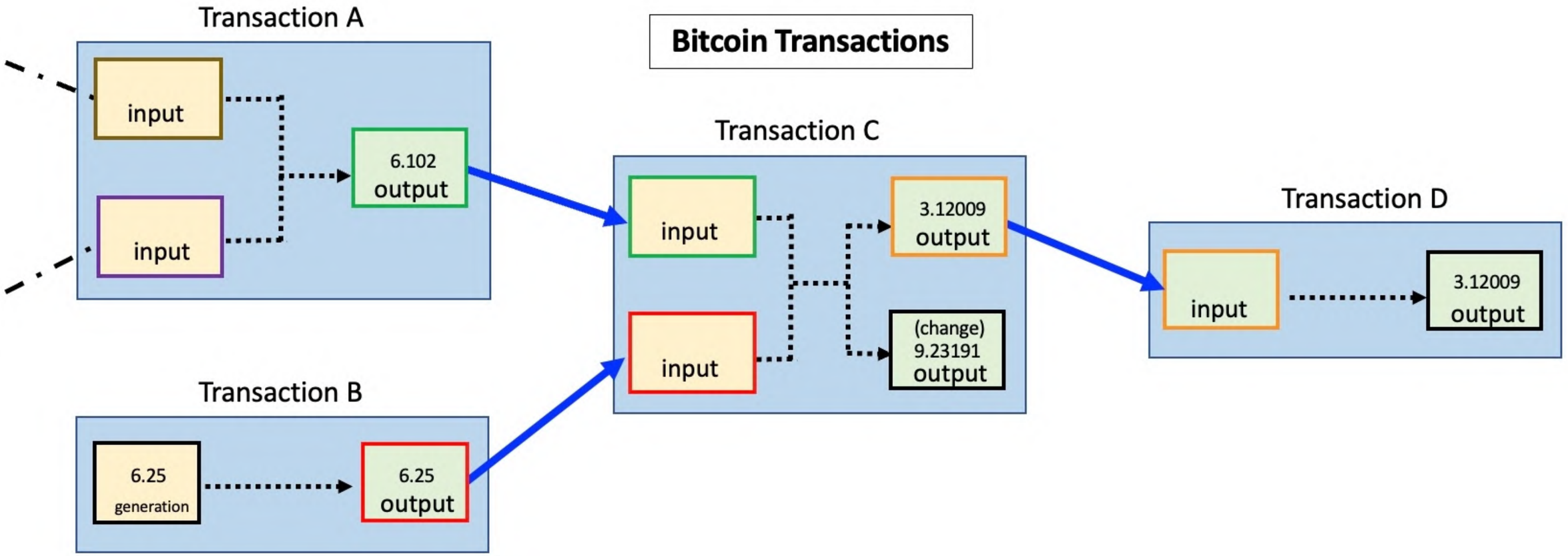
`3aba4162c7251c891207b747840551a71939b0de081f85c4e44cf7c13e41daa601`

Address

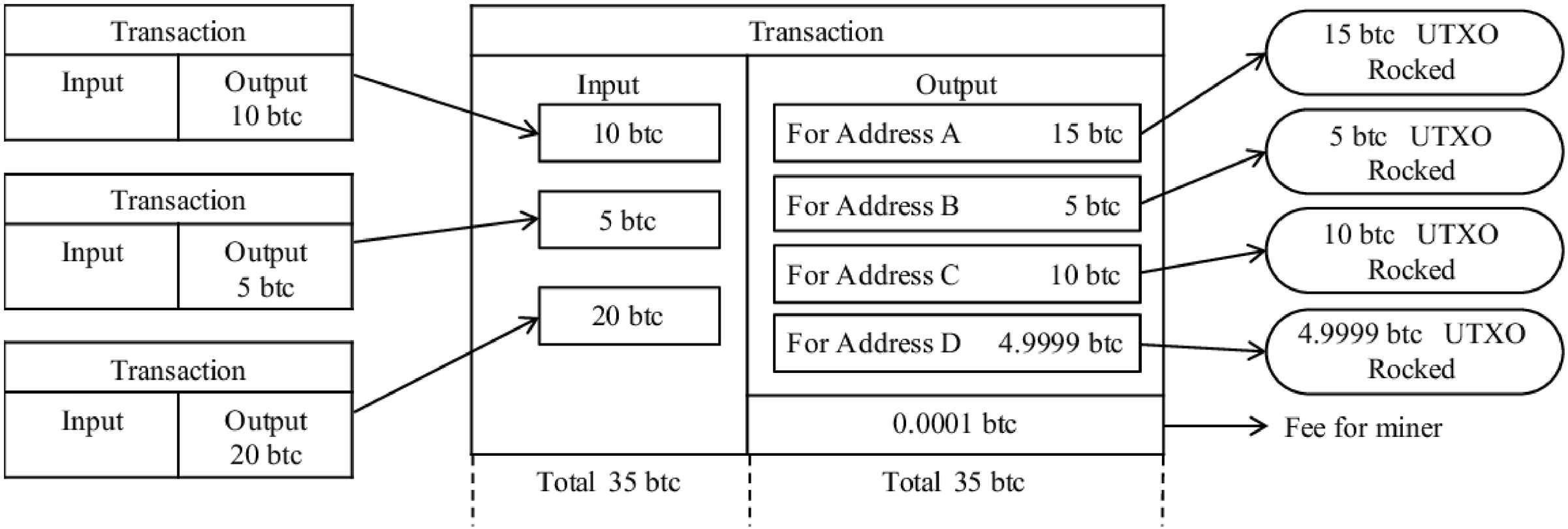
Address = F(Public Key)

`14cxpo3MBCYYWCgF74SWTdcmxipnGUsPw3`

Transactions



UTXO (Unspent Transaction Output)



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Thank you! Questions?

